

A Systematic Investigation of Lightweight Color Histogram Representations for Malaria Cell Classification

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Abstract—Automated malaria diagnosis has attracted considerable attention due to its potential to improve access to timely and accurate disease detection, particularly in regions where expert microscopy resources are limited. While deep learning methods have achieved impressive performance on malaria microscopy datasets, they often require substantial computational resources and large model sizes that may limit deployment in resource-constrained environments. This work investigates whether lightweight handcrafted color histogram features remain a competitive alternative for malaria cell classification.

Using the NIH Malaria Cell Images Dataset containing 27,558 segmented red blood cell images, we conduct a systematic study of color histogram representations and classical machine learning classifiers. The study evaluates RGB and HSV color spaces, performs channel-wise ablation of the Hue, Saturation, and Value components, compares five widely used classical classifiers, analyzes the influence of histogram resolution, and measures computational efficiency.

Experimental results demonstrate that HSV histograms consistently outperform RGB histograms, achieving a classification accuracy of 96.10%, an AUC of 0.9897, and a mean five-fold cross-validation accuracy of $96.06\% \pm 0.28\%$ using only a 24-dimensional feature vector and a Random Forest classifier. Channel ablation experiments reveal that the saturation component alone achieves 95.30% accuracy, whereas hue and value channels individually achieve approximately 76%, indicating that saturation distributions contribute most strongly to the discriminative capability of the proposed representation. Additional experiments show that increasing histogram resolution beyond eight bins provides only marginal performance improvements while substantially increasing feature dimensionality. The complete feature representation occupies only 2.52 MB of storage, and the final model performs inference in an average of 0.1857 ms per image.

These findings demonstrate that carefully engineered color histogram features, when combined with ensemble learning, provide an interpretable, computationally efficient, and highly accurate alternative to more complex learning frameworks for malaria cell classification. Furthermore, the systematic experimental analysis provides new insight into the relative importance of color representations and histogram design choices in lightweight medical image classification.

Index Terms—Malaria Detection, Medical Image Classification, Computer Vision, Machine Learning, Random Forest, HSV Color Space, Color Histograms, Feature Engineering, Pattern Recognition, Lightweight Artificial Intelligence

I. INTRODUCTION

Malaria remains one of the world’s most devastating infectious diseases, placing a significant burden on healthcare systems, particularly in low- and middle-income countries. According to the World Health Organization (WHO), hundreds of millions of malaria cases are reported annually, with the majority occurring in sub-Saharan Africa. Early and accurate diagnosis is essential for reducing mortality and ensuring timely treatment. Consequently, reliable automated diagnostic systems have become an active area of research within medical image analysis.

Microscopic examination of Giemsa-stained thin blood smears remains the clinical gold standard for malaria diagnosis. During this process, trained microscopists manually inspect red blood cells to determine the presence of malaria parasites. Although highly effective when performed by experienced personnel, manual microscopy is time-consuming, labor-intensive, and subject to inter-observer variability. In regions where trained experts and laboratory infrastructure are limited, these challenges motivate the development of computer-aided diagnostic systems capable of assisting clinical decision making.

Recent advances in artificial intelligence have substantially improved automated malaria detection. Deep convolutional neural networks (CNNs), including architectures such as ResNet, MobileNet, and EfficientNet, have achieved impressive classification performance on publicly available malaria microscopy datasets. However, these approaches typically require large numbers of trainable parameters, extensive computational resources, specialized hardware for training, and comparatively limited interpretability. Such requirements may restrict their practicality in resource-constrained settings where computational efficiency and model transparency are important considerations.

Classical machine learning approaches offer an alternative paradigm by combining handcrafted image representations with relatively lightweight classifiers. Compared with deep learning models, handcrafted feature methods generally require substantially smaller storage footprints, shorter training

times, and simpler deployment pipelines. Furthermore, they often provide greater interpretability because the extracted image characteristics can be directly analyzed. Despite these advantages, relatively little attention has been devoted to understanding which handcrafted color representations are most informative for malaria cell classification and how design choices such as color space, histogram resolution, and classifier selection influence overall performance.

Color information is particularly attractive for malaria microscopy because Giemsa staining alters the appearance of infected red blood cells in ways that may be captured through color distributions. While RGB histograms have traditionally been used as straightforward color descriptors, alternative color spaces such as HSV separate chromatic information from intensity information and may provide representations that better capture staining characteristics. Nevertheless, the relative contribution of individual HSV channels and the influence of histogram design remain insufficiently explored.

Rather than proposing a new classification algorithm, this work presents a systematic empirical investigation of lightweight color histogram representations for malaria cell classification. Using the publicly available NIH Malaria Cell Images Dataset, we evaluate multiple feature representations, compare several widely used classical machine learning algorithms, analyze the individual contributions of the Hue, Saturation, and Value channels, investigate the effect of histogram resolution, and assess computational efficiency in terms of storage requirements, training time, and inference speed. This experimental framework enables us to examine not only which representation performs best, but also why particular design choices lead to improved classification performance.

The principal contributions of this work are summarized as follows:

- We perform a comprehensive comparison of RGB and HSV histogram representations for malaria cell classification using identical experimental conditions and demonstrate that HSV consistently provides superior classification performance.
- We conduct a channel-wise ablation study that isolates the Hue, Saturation, and Value components individually, showing that saturation distributions contribute most strongly to the discriminative capability of the histogram representation.
- We compare five classical machine learning classifiers on the complete NIH Malaria Cell Images Dataset using identical HSV histogram features and identify Random Forests as the most effective classifier for this feature representation.
- We investigate the influence of histogram resolution by evaluating multiple bin configurations, demonstrating that increasing histogram granularity beyond eight bins provides only marginal performance improvements while substantially increasing feature dimensionality.
- We demonstrate that the proposed lightweight framework achieves 96.10% classification accuracy, a mean five-fold cross-validation accuracy of $96.06\% \pm 0.28\%$, an AUC

of 0.9897, and an average inference time of 0.1857 milliseconds per image using only a compact 24-dimensional feature vector occupying 2.52 MB of storage.

The remainder of this paper is organized as follows. Section II reviews related work in automated malaria detection and lightweight machine learning approaches. Section III describes the dataset, preprocessing pipeline, feature extraction methodology, and classification framework. Section IV presents the experimental design, while Section V discusses the experimental results. Section VI analyzes the findings and limitations of the study, and Section VII concludes the paper and outlines directions for future work.

II. RELATED WORK

Research on automated malaria diagnosis has evolved considerably over the past decade, progressing from handcrafted feature extraction techniques to sophisticated deep learning frameworks. The availability of publicly accessible microscopy datasets, particularly the NIH Malaria Cell Images Dataset, has accelerated the development and comparison of machine learning approaches for malaria parasite detection. Existing research can generally be categorized into three broad directions: traditional machine learning using handcrafted features, deep learning methods based on convolutional neural networks, and computationally efficient approaches designed for deployment in resource-constrained environments.

A. Traditional Machine Learning Approaches

Prior to the widespread adoption of deep learning, automated malaria detection primarily relied on handcrafted image features combined with conventional machine learning classifiers. These methods extracted manually designed descriptors representing characteristics such as color, texture, morphology, and statistical image properties before applying classifiers including Support Vector Machines (SVMs), Decision Trees, K-Nearest Neighbors (KNN), Logistic Regression, and Random Forests.

Handcrafted feature representations remain attractive because they are computationally inexpensive, interpretable, and generally require significantly less training data than deep neural networks. Furthermore, they provide direct insight into which image characteristics contribute most strongly to classification decisions, an advantage that is often unavailable in end-to-end deep learning systems.

Despite these advantages, relatively few studies have systematically investigated how different color representations influence classification performance. Most previous work selected a particular feature representation without examining whether alternative color spaces or individual color channels contributed differently to diagnostic performance.

B. Deep Learning for Malaria Classification

Recent research has largely shifted toward convolutional neural networks (CNNs), which automatically learn hierarchical feature representations directly from microscopy images. CNN-based models have consistently demonstrated excellent

classification performance on malaria microscopy datasets, frequently reporting accuracies exceeding 95%.

Transfer learning has become particularly popular due to its ability to leverage knowledge acquired from large natural image datasets. Architectures including ResNet, MobileNet, DenseNet, EfficientNet, and Inception have all been successfully adapted for malaria parasite classification. Boit and Patil employed EfficientNet-based architectures for malaria diagnosis, demonstrating that modern deep learning models can achieve high predictive performance while benefiting from transfer learning. Similarly, Mujahid *et al.* proposed an efficient deep learning framework that achieved strong classification performance using optimized convolutional neural network architectures.

Although these methods achieve impressive predictive accuracy, they generally require substantially larger computational resources than traditional machine learning approaches. Training often depends on GPUs, large model parameter counts, and extended optimization procedures. In addition, deep neural networks frequently provide limited interpretability, making it difficult to identify which visual characteristics contribute most strongly to classification decisions.

C. Computational Efficiency and Model Interpretability

The increasing interest in deploying artificial intelligence systems within resource-constrained healthcare environments has renewed attention toward lightweight machine learning methods. Compact models requiring minimal storage, rapid inference, and modest computational resources are particularly attractive for deployment on portable devices and low-cost computing platforms.

Random Forest classifiers represent an effective compromise between predictive performance and interpretability. Unlike many black-box models, Random Forests provide feature importance estimates that enable quantitative analysis of individual feature contributions. This capability allows researchers to investigate not only overall classification accuracy but also the relative importance of different image characteristics.

Similarly, histogram-based image descriptors remain appealing because they are simple to compute, highly compact, and computationally efficient. Their low dimensionality reduces both storage requirements and training complexity while preserving important statistical information about image color distributions.

D. Research Gap

While previous studies have demonstrated that both hand-crafted features and deep learning methods can successfully classify malaria microscopy images, comparatively little attention has been devoted to understanding how different color representations influence lightweight classification performance. Existing studies typically focus on maximizing classification accuracy using a single feature representation rather than systematically evaluating alternative color spaces, individual color channels, histogram design choices, and classifier behavior under identical experimental conditions.

Furthermore, the relative contribution of the Hue, Saturation, and Value components of the HSV color space has not been thoroughly investigated for malaria cell classification. Likewise, few studies have examined the relationship between histogram resolution and classification performance or quantified the computational tradeoffs associated with increasingly detailed histogram representations.

E. Position of This Work

Rather than proposing a new machine learning algorithm, this work presents a systematic experimental study of lightweight color histogram representations for malaria cell classification. We compare RGB and HSV histogram features under identical experimental conditions, evaluate multiple classical machine learning classifiers, isolate the contribution of individual HSV channels through channel-wise ablation, investigate the influence of histogram resolution, and analyze computational efficiency through storage, training time, and inference speed.

By emphasizing empirical analysis rather than algorithmic novelty, this work seeks to provide practical design guidance for researchers developing lightweight medical image classification systems and to improve understanding of how color representation choices influence malaria cell classification performance.

III. MATERIALS AND METHODS

This section describes the dataset, preprocessing pipeline, feature extraction methods, classification models, and evaluation procedures used throughout this study. All experiments were conducted under identical conditions unless otherwise specified to ensure fair comparison among feature representations and classifiers.

A. Dataset

Experiments were performed using the publicly available NIH Malaria Cell Images Dataset introduced by Rajaraman *et al.* The dataset contains segmented red blood cell (RBC) images extracted from thin blood smear slides collected from malaria-infected patients.

The dataset consists of a total of 27,558 RGB images equally divided into two classes:

- 13,779 parasitized (infected) cells
- 13,779 uninfected cells

Because the dataset is perfectly balanced, overall classification accuracy provides a meaningful evaluation metric without requiring additional balancing techniques.

Each image contains a single segmented red blood cell and serves as one independent sample for binary classification.

B. Image Preprocessing

All images were resized to a fixed resolution of 64×64 pixels to ensure consistent feature extraction while reducing computational complexity.

Images were loaded using OpenCV in BGR format and immediately converted to RGB color space. No additional

preprocessing operations such as segmentation, denoising, illumination correction, contrast enhancement, or data augmentation were applied. Consequently, the extracted histogram features describe the original color distribution of each resized cell image.

C. Color Histogram Feature Extraction

The objective of this work is to investigate lightweight color representations for malaria cell classification. Histogram-based features were selected because they provide compact statistical descriptions of image color distributions while remaining computationally inexpensive to compute and store.

Two color spaces were investigated: RGB and HSV.

1) *RGB Histogram Representation*: For the RGB representation, each image was separated into its Red, Green, and Blue channels.

An eight-bin normalized histogram was computed independently for each channel over the intensity range $[0, 256]$.

The three normalized histograms were concatenated to form a single feature vector.

Consequently, the RGB representation contains

$$3 \times 8 = 24$$

features per image.

2) *HSV Histogram Representation*: To investigate whether an alternative color representation better captures malaria-related staining characteristics, images were converted from RGB to HSV color space.

Separate normalized histograms were computed for the Hue, Saturation, and Value channels using the same histogram construction procedure employed for RGB.

Unless otherwise stated, each channel was represented using eight histogram bins, resulting in a 24-dimensional feature vector.

3) *Channel Ablation*: To better understand the contribution of individual HSV components, additional feature representations were constructed using each channel independently.

Four HSV-based representations were therefore evaluated:

- Hue only (H)
- Saturation only (S)
- Value only (V)
- Complete HSV representation

These experiments isolate the discriminative contribution of each channel while maintaining identical classifier and evaluation settings.

4) *Histogram Resolution Study*: Histogram resolution determines the granularity with which color distributions are represented.

To investigate its influence on classification performance, four histogram resolutions were evaluated:

- 4 bins per channel
- 8 bins per channel
- 16 bins per channel
- 32 bins per channel

These correspond to feature vectors containing 12, 24, 48, and 96 dimensions, respectively.

All remaining experimental settings were kept identical.

D. Machine Learning Models

Five widely used classical machine learning algorithms were evaluated.

- K-Nearest Neighbors (KNN)
- Logistic Regression
- Support Vector Machine (RBF kernel)
- Gaussian Naive Bayes
- Random Forest

The purpose of this comparison was to determine which classifier most effectively exploits compact color histogram features.

Random Forest consistently achieved the strongest predictive performance and was therefore selected as the primary classifier for all subsequent feature representation experiments.

The final Random Forest model consisted of 200 decision trees with bootstrap aggregation and a fixed random seed to ensure reproducibility.

E. Experimental Setup

For all experiments, the dataset was randomly partitioned into training and testing subsets using an 80/20 split.

The same train-test partition was maintained throughout all feature representation comparisons to ensure fair evaluation.

Unless otherwise specified, all reported results correspond to the Random Forest classifier trained on the selected feature representation.

F. Performance Evaluation

Model performance was evaluated using several complementary metrics.

Classification performance was measured using:

- Accuracy
- Precision
- Recall
- F1-score
- Sensitivity
- Specificity
- Area Under the ROC Curve (AUC)

To assess robustness and reduce dependence on a single train-test partition, five-fold cross-validation was also performed on the complete dataset.

Finally, computational efficiency was evaluated using:

- Training time
- Average inference time per image
- Feature dimensionality
- Storage size of the extracted feature representation

Together, these metrics provide a comprehensive evaluation of both predictive performance and computational efficiency.

IV. EXPERIMENTAL RESULTS

This section presents the experimental evaluation of the proposed lightweight color histogram framework. The experiments were designed to answer five research questions concerning classifier selection, feature representation, channel contribution, histogram design, and computational efficiency.

A. Experiment 1: Which Classical Machine Learning Classifier Performs Best?

The first experiment compares five widely used classical machine learning algorithms using identical HSV histogram features extracted from the complete NIH Malaria Cell Images Dataset. All classifiers were trained and evaluated using the same train-test partition to ensure a fair comparison.

Table I summarizes the results.

TABLE I
COMPARISON OF CLASSICAL MACHINE LEARNING CLASSIFIERS USING HSV HISTOGRAM FEATURES

Classifier	Accuracy (%)	Training Time (s)
K-Nearest Neighbors	75.44	0.01
Logistic Regression	68.32	1.45
Support Vector Machine (RBF)	70.63	21.91
Gaussian Naive Bayes	70.08	0.01
Random Forest	96.10	9.88

Random Forest substantially outperformed all competing classifiers, achieving more than twenty percentage points higher classification accuracy than the nearest alternative. Interestingly, both Logistic Regression and Support Vector Machines performed considerably worse despite the relatively low dimensionality of the feature representation. This suggests that the relationships encoded within the histogram features are highly nonlinear and are more effectively modeled by ensemble learning methods than by linear or distance-based classifiers.

Based on these findings, Random Forest was selected as the primary classifier for all remaining experiments.

B. Experiment 2: Does HSV Outperform RGB?

Having identified Random Forest as the most effective classifier, the second experiment investigates whether HSV provides a more informative color representation than the conventional RGB color space.

Both representations use identical preprocessing, identical train-test partitions, and identical Random Forest parameters. Each representation contains 24 features generated from eight normalized histogram bins per color channel.

Table II presents the comparison.

TABLE II
COMPARISON OF RGB AND HSV HISTOGRAM REPRESENTATIONS

Representation	Accuracy (%)	Training Time (s)
RGB Histogram	92.42	27.49
HSV Histogram	96.10	16.53

HSV histograms consistently outperformed RGB histograms while simultaneously reducing training time. The

improvement of approximately 3.7 percentage points demonstrates that separating chromatic information into Hue, Saturation, and Value components provides a more discriminative representation than directly modeling Red, Green, and Blue intensity distributions.

C. Experiment 3: Which HSV Channel Contributes Most?

Although HSV substantially improved classification performance, it remained unclear which channel contributed most strongly to this improvement. To answer this question, channel-wise ablation experiments were performed using the Hue, Saturation, and Value channels individually before evaluating the complete HSV representation. The results are summarized in Table III.

TABLE III
HSV CHANNEL ABLATION STUDY

Representation	Accuracy (%)	AUC	CV Accuracy (%)
Hue (H)	75.93	0.8484	75.11 $\pm$ 0.25
Value (V)	76.25	0.8521	76.43 $\pm$ 0.79
Saturation (S)	95.30	0.9838	95.55 $\pm$ 0.23
HSV	96.10	0.9897	96.06 $\pm$ 0.28

The results reveal a striking disparity among the three HSV channels. When evaluated independently, both Hue and Value achieve classification accuracies of approximately 76%, whereas the Saturation channel alone reaches 95.30% accuracy. Incorporating Hue and Value into the complete HSV representation provides a modest additional improvement to 96.10%.

These findings indicate that, within the investigated histogram representations, saturation distributions contain the majority of the discriminative information required for malaria cell classification, while Hue and Value contribute complementary information that further refines the final decision boundary.

Figure 1 presents the confusion matrix for the best-performing model, confirming strong and balanced classification performance across both classes.

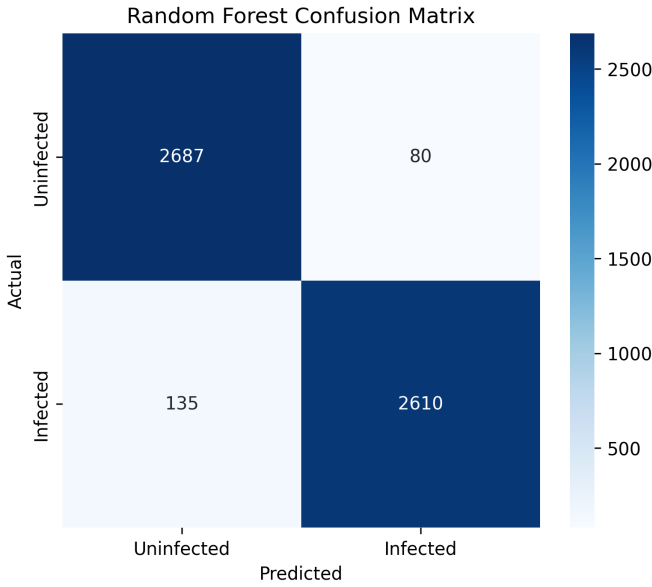


Fig. 1. Confusion matrix for the HSV histogram and Random Forest classifier. The model correctly classifies 2,687 of 2,767 uninfected cells and 2,610 of 2,745 infected cells, with false negative and false positive rates of 4.9% and 2.9%, respectively.

D. Experiment 4: How Many Histogram Bins Are Necessary?

The fourth experiment investigates the influence of histogram resolution on classification performance.

Increasing the number of histogram bins provides a more detailed representation of color distributions but also increases feature dimensionality and computational cost.

Four histogram resolutions were evaluated while maintaining identical Random Forest parameters.

The results are presented in Table IV.

TABLE IV
INFLUENCE OF HISTOGRAM RESOLUTION

Bins	Features	Accuracy (%)	AUC	CV Accuracy (%)
4	12	94.94	0.9850	94.62 ± 0.23
8	24	96.10	0.9897	96.06 ± 0.28
16	48	96.15	0.9910	96.21 ± 0.24
32	96	96.30	0.9906	96.17 ± 0.25

Increasing histogram resolution from four to eight bins substantially improves classification performance. However, additional increases to sixteen and thirty-two bins produce only marginal gains despite doubling and quadrupling the feature dimensionality.

Consequently, the eight-bin representation provides the best balance between predictive performance, computational efficiency, and feature compactness. This configuration was therefore selected as the proposed representation throughout the remainder of the study.

E. Experiment 5: Computational Efficiency

In addition to predictive performance, practical deployment requires computational efficiency.

TABLE V
COMPUTATIONAL EFFICIENCY OF THE PROPOSED METHOD

Metric	Value
Feature Dimension	24
Storage Size	2.52 MB
Training Time	16.53 s
Inference Time	0.1857 ms/image
Images per Second	≈ 5385

Table V summarizes the computational characteristics of the proposed eight-bin HSV representation.

The complete feature representation for all 27,558 cell images occupies only 2.52 MB of storage while maintaining excellent predictive performance. Furthermore, the trained Random Forest model performs inference in approximately 0.186 milliseconds per image, corresponding to a throughput exceeding 5,000 images per second on the evaluation hardware.

These results demonstrate that the proposed framework provides an attractive combination of accuracy, interpretability, compactness, and computational efficiency.

F. Summary of Experimental Findings

Taken together, the experimental results reveal several important observations.

First, Random Forest substantially outperforms all other evaluated classical machine learning algorithms when applied to compact color histogram features. Second, HSV provides a considerably more informative feature representation than RGB. Third, channel-wise ablation demonstrates that saturation distributions account for the majority of the observed performance improvement. Fourth, increasing histogram resolution beyond eight bins yields diminishing returns despite substantially increasing feature dimensionality. Finally, the proposed framework combines high predictive performance with low storage requirements and extremely rapid inference, making it well suited for lightweight deployment scenarios.

V. DISCUSSION

The objective of this study was not to propose a new machine learning algorithm, but rather to investigate how lightweight color histogram representations influence malaria cell classification performance. Through a series of controlled experiments, several important observations emerged regarding feature representation, classifier behavior, histogram design, and computational efficiency.

A. Effectiveness of HSV Color Representation

The first major finding of this work is that HSV provides a substantially more discriminative representation than the conventional RGB color space.

Using identical preprocessing, identical feature dimensionality, identical train-test partitions, and the same Random Forest classifier, HSV histograms improved classification accuracy from 92.42% to 96.10% while simultaneously reducing training time.

This result suggests that separating chromatic information into Hue, Saturation, and Value components captures malaria-related staining characteristics more effectively than directly modeling RGB intensity distributions. Unlike RGB, the HSV representation partially decouples color information from brightness, allowing the classifier to exploit variations that may be less apparent in the original color space.

B. Importance of Saturation

Perhaps the most significant finding of this study arises from the channel ablation experiments.

Evaluating each HSV channel independently revealed a striking difference in predictive capability. While both the Hue and Value channels achieved classification accuracies of approximately 76%, the Saturation channel alone achieved 95.30% accuracy, only slightly below the complete HSV representation.

Figure 2 illustrates the average saturation histogram distributions for infected and uninfected cells.

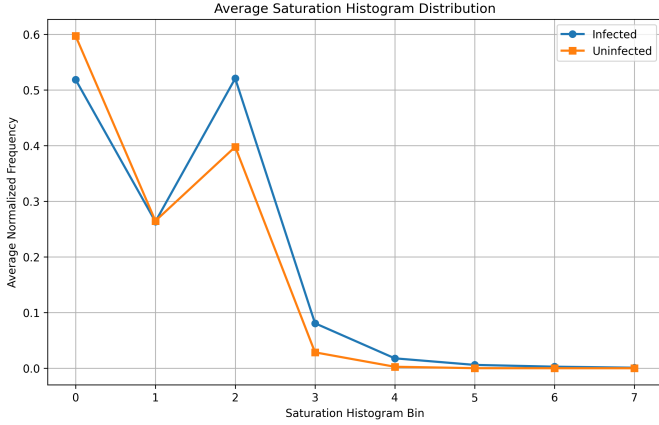


Fig. 2. Average normalized saturation histograms for parasitized and uninfected cells.

Although the average histogram distributions appear visually similar, consistent differences exist across several histogram bins. The Random Forest classifier exploits these subtle differences collectively, allowing saturation alone to provide remarkably strong classification performance.

This observation suggests that, within the investigated histogram representations, variations in staining intensity contribute substantially more discriminative information than hue or brightness alone. Importantly, this conclusion is limited to histogram-based color representations and should not be interpreted as a biological claim regarding malaria pathology.

C. Classifier Behavior

The comparison among classical machine learning algorithms demonstrates that classifier selection has a profound impact on performance when using compact histogram features.

Random Forest achieved 96.10% accuracy, whereas all remaining classifiers produced accuracies between approximately 68% and 75%.

Such a large performance gap indicates that the relationships among histogram features are highly nonlinear. Ensemble learning methods are able to capture complex interactions among histogram bins that are not adequately modeled by linear classifiers or distance-based methods.

An additional advantage of Random Forest is its ability to estimate feature importance.

Figure 3 presents the feature importance scores obtained from the final model.

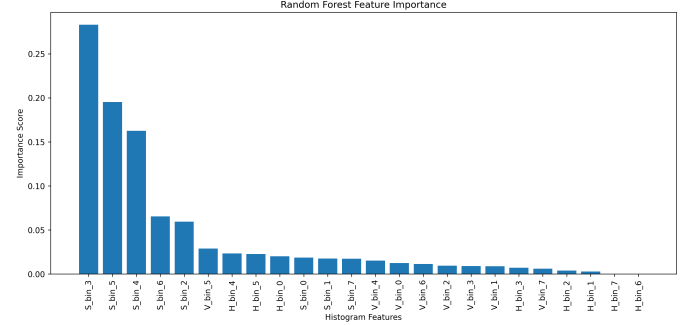


Fig. 3. Feature importance obtained from the Random Forest classifier.

The feature importance analysis supports the conclusions drawn from the channel ablation experiments by identifying saturation-related histogram bins as the dominant contributors to classification performance.

D. Histogram Resolution Tradeoff

A second objective of this work was to determine an appropriate histogram resolution for lightweight malaria classification.

Increasing histogram resolution from four bins to eight bins produced a noticeable improvement in classification performance. However, additional increases to sixteen and thirty-two bins yielded only marginal improvements despite substantially increasing feature dimensionality.

This behavior suggests diminishing returns beyond eight histogram bins. Although the thirty-two-bin representation achieved the highest measured accuracy, the improvement relative to eight bins was only 0.20 percentage points while requiring four times as many features.

Consequently, the eight-bin representation represents an effective engineering tradeoff between predictive performance, storage requirements, and computational complexity.

E. Computational Efficiency

Lightweight machine learning methods remain particularly attractive for deployment in environments where computational resources are limited.

The proposed framework represents each cell using only 24 numerical features, requiring a total storage footprint of approximately 2.52 MB for the complete NIH dataset.

Furthermore, the trained Random Forest classifier performs inference in approximately 0.1857 milliseconds per image,

corresponding to a throughput exceeding 5,000 images per second.

These computational characteristics demonstrate that high classification performance does not necessarily require computationally intensive deep learning architectures. Instead, carefully engineered feature representations combined with ensemble learning can provide an attractive balance between efficiency and predictive accuracy.

F. Comparison with Existing Approaches

A quantitative comparison with previously published methods on the NIH Malaria Cell Images Dataset, including deep learning approaches, is currently in preparation and will be included in a subsequent version of this manuscript.

VI. LIMITATIONS

Several limitations should be acknowledged.

First, all experiments were conducted using the NIH Malaria Cell Images Dataset. Although this dataset is widely recognized as a benchmark for malaria classification, additional validation on independently collected datasets would provide stronger evidence regarding the generalizability of the proposed framework.

Second, histogram features summarize global color distributions but do not explicitly encode spatial information, texture, or cell morphology. Consequently, potentially informative structural characteristics of infected cells are not directly represented.

Third, this study focuses exclusively on handcrafted histogram representations and classical machine learning algorithms. While comparisons with modern deep learning architectures are discussed using previously published results, identical experimental comparisons were beyond the scope of this work.

Finally, although feature importance analysis and channel ablation provide useful interpretability, the observed dominance of saturation-based features should be interpreted as a property of the investigated histogram representation rather than definitive evidence regarding the biological characteristics of malaria infection.

VII. CONCLUSION

This paper presented a systematic investigation of lightweight color histogram representations for malaria cell classification.

Through a comprehensive series of experiments, we compared RGB and HSV color spaces, evaluated multiple classical machine learning classifiers, isolated the contribution of individual HSV channels, investigated histogram resolution, and analyzed computational efficiency.

The experimental results demonstrate that HSV histograms consistently outperform RGB histograms and that saturation distributions account for the majority of the observed performance improvement. Furthermore, Random Forest classifiers substantially outperform alternative classical machine learning methods when applied to compact color histogram features.

Using only a 24-dimensional feature vector, the proposed framework achieved 96.10% classification accuracy, a mean five-fold cross-validation accuracy of $96.06\% \pm 0.28\%$, an AUC of 0.9897, and an average inference time of 0.1857 milliseconds per image while requiring only 2.52 MB to represent the complete dataset.

Beyond the proposed classification framework, this study provides empirical insight into the influence of color representation, channel selection, histogram resolution, and classifier choice in lightweight medical image classification. These findings may serve as practical design guidance for future research on computationally efficient computer-aided diagnostic systems.

Future work will focus on validating the proposed framework on additional malaria microscopy datasets, investigating robustness across different acquisition conditions and parasite species, and exploring hybrid lightweight representations that combine compact color descriptors with complementary texture features.

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